2024 4th Int. Conf. on Innovations in Science, Engineering and Technology (ICISET)
26-27 October 2024, Chittagong, Bangladesh

2024 International Conference on Innovations in Science, Engineering and Technology (ICISET) | 979-8-3503-5549-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICISET62123.2024.10939797

IoT Based Smart Walking Stick for Enhanced Mobility of The Visually Impaired

Md Mehedi Hassain
Dept. of Electrical and Electronic Engineering
International Islamic University Chittagong
Chittagong 4314, Bangladesh
mdmehedihasan225588@gmail.com

***Abstract*—Vision is a crucial aspect of human life, but unfortunately, some people are visually impaired and face significant daily challenges. This issue intensifies when navigating unfamiliar areas. Blindness still lacks adequate global support, despite numerous efforts to improve mobility for the visually impaired. Traditional white canes remain basic tools, and while various electronic aids are available, they often suffer from high costs, limited user-friendliness, and accuracy issues. Our research introduces an IoT-based Smart Walking Stick designed to enhance self-sufficiency for blind individuals. This system provides real-time monitoring and safety alerts through pre-recorded voice messages, ensuring user security. It integrates several components, including the ESP32, ultrasonic sensors for obstacle detection in front, left, and right directions, IR sensors for detecting road obstacles, a water level sensor for detecting the probability of water in potholes, and a mini solar panel for battery backup. Pre-recorded voice messages, stored on an SD card, offer auditory alerts to the user. Additionally, GPS and GSM modules facilitate live location tracking and SMS alerts to guardians. The ESP32 processes and transmits sensor data to a Node.js server via MQTT, enabling guardians to monitor the user's location and status through a dedicated website or mobile app. This smart stick is crafted to empower visually impaired users by improving their independence and comfort through an affordable and innovative solution.**

***Index Terms*—Walking Stick Monitoring system, IOT, Ultrasonic Sensor, IR sensor, Water level sensor, Solar panel, GPS and GSM alert, Website or Mobile app, MQTT communication protocol.**

I. Introduction

Bangladesh, despite its poverty, ranks among the world's highest in blindness rates. Many of its citizens face partial or complete blindness. Over 750,000 individuals are registered as blind by the Department of Social Services, with an additional six million facing visual impairments. In Bangladesh, 0.59 percent of the population is blind, in contrast to 0.9 percent in the USA, according to recent reports [1]. Several previous studies have investigated this issue.

In this study, a Blind Stick with ultrasonic and infrared sensors detects obstacles up to 150 cm away, including those above knee level, ensuring safety both indoors and outdoors. It features staircase detection, alerting users to potential hazards. family members can track the Blind person location via a smartphone app [2]. In this study, an IoT-based Smart Shoe for the blind uses ultrasonic sensors and an Arduino UNO board. Equipped with multiple sensors, a micro-controller, and buzzers, the shoe alerts the wearer to obstacles with a buzz. Additionally, IoT-enabled Smart Glasses, equipped with sensors, enhance object detection over a wider area. Paired with Smart Shoes, they communicate to prevent obstacles, ensuring safe navigation for the Blind person [3]. In this research, the Smart Stick utilizes an ESP32, ultrasonic sensor, GPS receiver, GSM module, and speech playback. The Smart Stick detects obstacles and provides vocal alerts. It also calculates proximity, locates the blind user, and sends updates to family members [4]. In this study, a Raspberry Pi with a camera captures live video sent to cloud storage via a private blockchain. Deep learning servers analyze footage, sending obstacle alerts to the user, who receives audio notifications through an earphone. With an RTX 3090 GPU, the system provides notifications within 2 seconds, offering swift assistance to the visually impaired [5]. A smart blind stick for blind people equipped with ultrasonic, light, water, and height sensors for obstacle detection and environmental awareness. With GPS for location tracking and a voice recognition system for communication, this adjustable-height stick offers unparalleled support for the visually impaired [6]. This smart stick for blind people incorporates ultrasonic sensors, a speaker for alerts, and enables emergency location sharing via a mobile app. Utilizing Arduino Uno, Bluetooth, and a gyroscope, this stick ensures safety and real-time communication. It provides voice and vibration alerts for obstacles, with automated emergency activation for added security [7]. In this research, multiple sensors are integrated into the individual's stick, enabling it to detect obstacles and vital signs. It communicates with caregivers through a connected app, while GPS tracking ensures blind individuals using the stick can track their location accurately [8]. This smart stick integrates obstacle identification, GPS-based detection, and IoT connectivity for seamless daily tasks. Using ultrasonic and sensor technology, it identifies obstacles and patterns. GPS and IoT pinpoint the user's location and activate alarm mode if needed [9].In this study, a Smart Blind Stick with ultrasonic sensing, camera vision, and a speaker system detects obstacles and alerts users seamlessly. With emergency GSM and GPS features,

979-8-3503-5549-9/24/$31.00 ©2024 IEEE
Authorized licensed use limited to: VIT University. Downloaded on October 29,2026 at 03:22:45 UTC from IEEE Xplore. Restrictions apply.

it ensures safety by sending precise location alerts via SMS [10]. The IoT-based Smart Sense Cane was designed using SolidWorks 3D modeling software. Its ultrasonic sensors detect water, obstacles, and fire, while seamlessly guiding visually impaired individuals up and down stairs. Controlled by Arduino, it alerts users to hazards and syncs their location with family via Blynk, ensuring safety and peace of mind [11]. In this study, the Smart Sense Belt, equipped with two ultrasonic sensors and a Raspberry Pi, detects obstacles and holes, delivering real-time audio alerts. The system has been designed to accommodate gait movement as well as the variability of flat surfaces. Convenient and wearable, it outperforms traditional methods like the white stick, activating seamlessly upon wear and auto-calibrating for immediate assistance [12]. A smart stick equipped with ultrasonic, water, and LDR sensors, along with a vibrator motor, offers real-time environmental feedback. Integrated with Arduino Uno, it delivers precise signals to buzzers, vibrators, and voice alarms, ensuring prompt alerts. GPS navigation in the mobile can be used to guide the blind to new and unfamiliar places [13]. The system, equipped with ultrasonic sensors to detect obstacles, a water sensor to warn of wet surfaces, and a vibrating motor and buzzer to provide immediate feedback, ensures enhanced safety and independence. With the Arduino Mega integrated into the stick, obstacles are swiftly identified as the user moves, facilitating a rapid response [14]. In this study, ultrasonic sensors and a buzzer are utilized to detect obstacles and provide immediate feedback, enhancing mobility. Additionally, a stereoscopic sonar system and vibrators detect nearby obstacles, offering precise location feedback. Moreover, a remote-activated buzzer aids in stick retrieval for the visually impaired, enhancing accessibility and safety for the blind [15]. In this study, a smart cane is developed to detect elevated and depressed surfaces, aid stair navigation, and identify overhead obstacles. Equipped with moisture and ultrasonic sensors interfaced with Arduino, it offers real-time feedback to prevent falls and alerts users with voice output [16].

Previous research has focused on several key areas: obstacle detection, GPS location tracking, and buzzer alerts. Our study combines sonar technology with ultrasonic sensors to identify obstructions in the left, right, and front directions. In addition, IR sensors have been utilised to detect obstacles on the road, and a water level sensor is used to detect potholes that have accumulated water in them. Alerts are delivered by prerecorded voice messages that are saved on an SD card reader, which helps the visually handicapped navigate on-site. Additionally, guardians receive real-time location alerts via SMS from GPS and GSM modules, which are powered by a tiny solar panel to charge Li-ion batteries. The data accumulator is the ESP32, which communicates with the Node.js server via the MQTT communication protocol. A simple web page or mobile application allows guardians to keep updated on the location and overall condition of visually impaired individuals. With only a single click, guardians can locate the exact location of the individual with this platform's marker based display.

II. Overview of the System

The system, depicted in Fig. 1, integrates ESP32, GPS NEO-6MV2, Ultrasonic sensor, IR sensor, Water level sensor, SD card reader, speaker, and SIM800L GSM. ESP32 employs MQTT communication for data exchange. It monitors road obstacles and pothole water levels using ultrasonic and IR sensors, respectively. Pre-recorded voice alerts and GSM notifications ensure road safety. GPS locates the blind person, while web and mobile apps offer graphical monitoring. Solar panels ensure continuous operation.

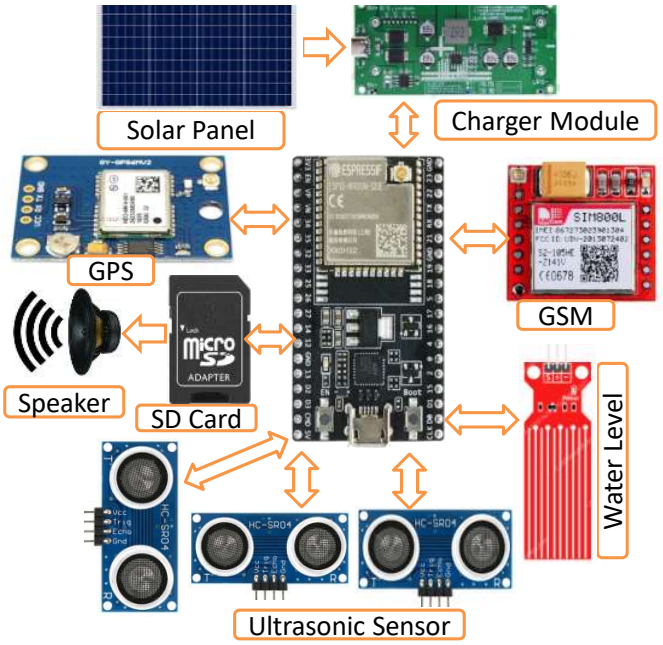


Fig. 1: Block Diagram of the System.

III. METHODOLOGY

This section outlines the way the prototype of the system developed. The Smart Walking Stick is being controlled, obstacles are being detected, path hole water is detectable, recorded voice alarms are being sent, the location of blind people is being read, and SMS messages are being sent using GPS and GSM modules. Additionally, the blind person's current location and status has been transmitted to the web server in real time.

A. Circuit Diagram of the System

In Fig. 3, the ESP32, SIM800L, and GPS NEO-6MV2 are connected via TX and RX pins, while the Ultrasonic Sensors' Trig and Echo pins are linked to the ESP32 using D12, D13, D14, D27, D32, and D33. The IR Sensor output pin D4 is connected to the ESP32. The Water Level Sensor D2 pin is connected to the ESP32. The SD Card reader pins D5, D18, D19, and D23 are connected to the ESP32. The Solar Panel is connected to the Battery Charging Module's input positive

Authorized licensed use limited to: VIT University. Downloaded on October 29,2026 at 03:22:45 UTC from IEEE Xplore. Restrictions apply.

and negative pins, while the Current Regulator and Li-ion battery are connected to the Battery Charging Module's pins 3, 4, 5, and 6. Utilizing the MQTT communication protocol, the ESP32 efficiently collects and transmits data to the server.

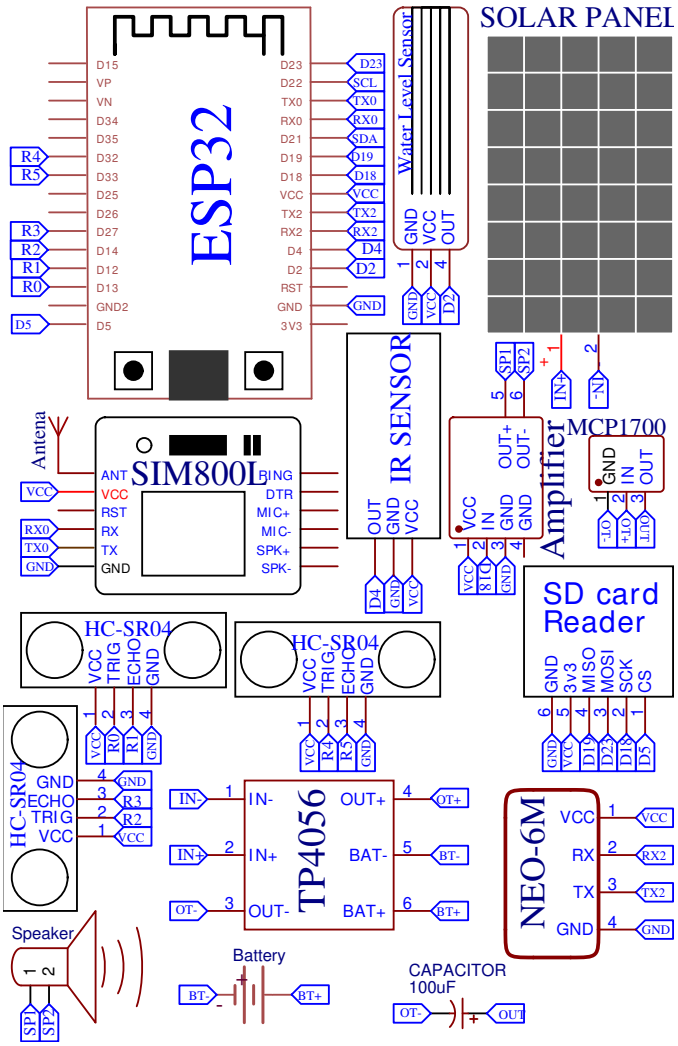


Fig. 2: Circuit Diagram of the System.

B. Data Transmission and Website Data Storage

The MQTT protocol was employed for data transfer to the cloud. In Fig. 3 illustrates the MQTT protocol's functionality. The web server frontend has been developed using HTML5, CSS3, JavaScript, and Chart.js. Real-time web servers are operated using Node.js, and the backend is a combination of Node.js and a MySQL database. The ESP32 gathers all sensor data and sends it to the Node.js server using the MQTT communication protocol. The MQTT protocol, showcasing its two primary client groups: publishers and subscribers. Unlike traditional client server architecture, where clients communicate directly with endpoints, MQTT clients interact with a broker. Publishers send data, while subscribers receive it based on predefined topics, enabling efficient message filtering and delivery. For authentication, we employed the TOKEN as both the username and password for MQTT, ensuring secure data transmission. MySQL ensures scalability, enabling effective handling of large volumes of data.

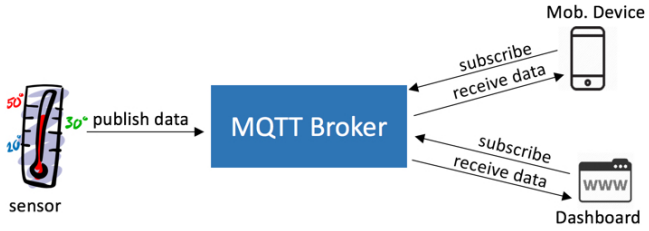


Fig. 3: MQTT(MQ Telemetry Transport) Process

C. Reading Location and Sending alert SMS

The guardian of the blind individual receives an SMS from the framework containing the person's current condition. To get the location and transmit the SMS, respectively, the GPS NEO-6MV2 and GSM (SIM800L) modules are utilised. The microcontroller ESP32 communicates with the GSM via a few AT commands and the UART (UniversalAsynchronous Receiver and Transmitter) communication protocol. The baud rate used in serial communication is 115200 bps. A list of AT commands for sending and receiving SMS messages as well as reading locations is included in "Table I" [17].

TABLE I: AT COMMAND LIST

AT Commands	Functions
AT+CPIN	Returns READY if the modem is connected to the network
AT+CMGF=1	Establishing SMS as a text message
AT+CMGS="Cell-in Number" <Enter> Body of Message <Ctrl-Z>	Sending an SMS
AT+CNMI=2,2,0,0,0	Activating incoming SMS direct reading
AT+SAPBR=3,1,"Contype","GPRS"	Set GPRS as the connection type
AT+SAPBR=3,1,"APN","www"	Activate GPRS
AT+SAPBR=2,1	Query If the connection is configured correctly and we receive a response with an IP address, we can continue
AT+CIPGSMLOC=1,1	Location (latitude and longitude)

D. Real Time Blind people Location Track

Graphical view on the website showing the blind person's path with location markers is shown in Fig. 4, This project stands out as it capitalizes on of a frontend which

Authorized licensed use limited to: VIT University. Downloaded on October 29,2026 at 03:22:45 UTC from IEEE Xplore. Restrictions apply.

was developed using HTML5, CSS3, and JavaScript and leverages Chart.js' dynamic visualization features. Real-time web servers are operated by Node.js with a backend, enabling to enable seamless combination of MySQL databases with scalable network applications. The integration system ensures an accurate portrayal of the blind person path locations on the website.

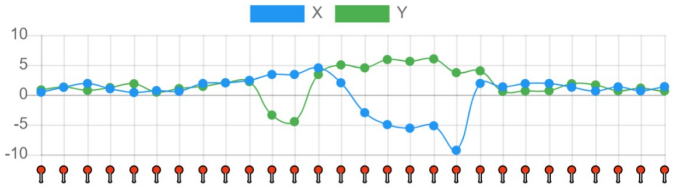


Fig. 4: Graphical View on website with blind people path with location marker.

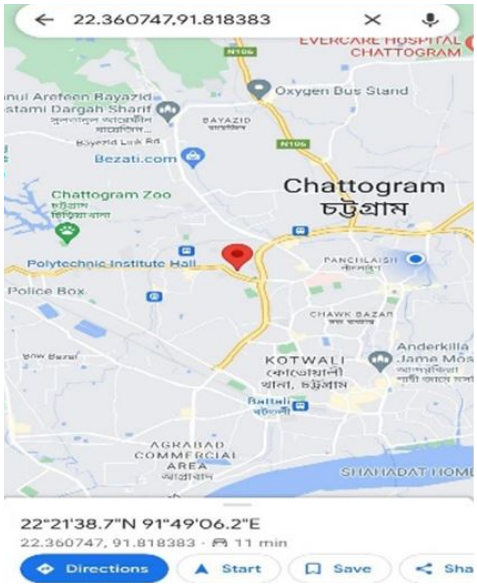


Fig. 5: Exact location of the blind person on Google Maps.

The exact positioning of the blind person road location on Google Maps appears in Fig. 5. The blind person's real time Google Map location can be precisely displayed by clicking on the server's graphical location pin. The proposed idea features an innovative graphical user interface for navigation positioning. The key objective is to develop an accurate graphical monitoring system.

E. Obstacle Detection

Ultrasonic sensors detect road obstacles. Fig. 6 illustrates the functionality of the ultrasonic sensor. Sonar sensors are used to detect obstacles in the left, right, and front directions. The microprocessor transitions the trigger pin of the sonar sensor from low to high. The sonar sensor generates and sends out eight 42 KHz sound pulses in response. The echo pin on the sonar sensor turns HIGH during this period. The echo pin's level decreases if the

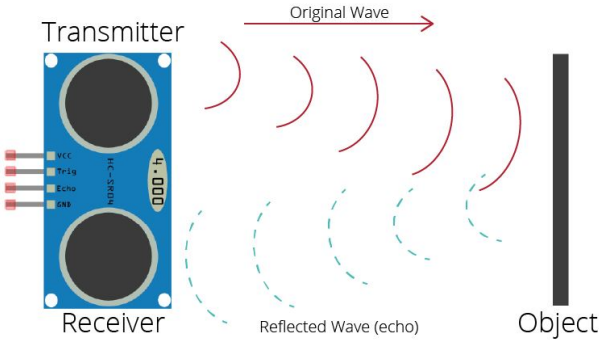


Fig. 6: Ultrasonic Sensor Working Process

sonar receiver receives a sound signal from the sensor. The duration, shown as "D," which stands for the obstacle's distance, is equal to the length of the HIGH signal on the echo pin. Equation 1 is used by the ultrasonic sensor to determine the distance to the obstacle and, from there, to calculate the time required for the signal to be reflected at specific distances.

Distance (D) = 0.028T + 1.093 (1)

T is the expected time in this instance. "T" represents the microcontroller's accurate time measurement in microseconds, and "D" represents the distance in centimetres in this instance. The fundamental computation of obstacle distances is provided by this equation. The relationship between object detection by Ultrasonic Sensors and operation time is displayed in "Table II" [18].

TABLE II: OPERATING TIME IDENTITY OF ULTRASONIC SENSOR OBJECT

Distance (cm)	Average Time (ms)
5	139.54
15	496.68
25	853.83
35	1210.97
45	1568.11
55	1925.25
65	2282.39
75	2639.54

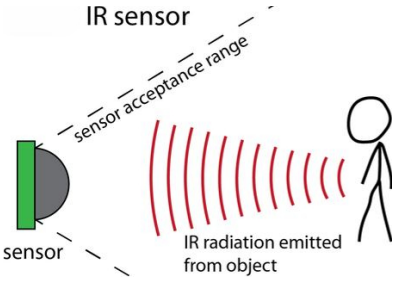


Fig. 7: IR Sensor Working Process

In Fig. 7 demonstrates the working process of the IR Sensor. By generating infrared waves that strike things and reflect signals back to the sensor, the IR Sensor is used

Authorized licensed use limited to: VIT University. Downloaded on October 29,2026 at 03:22:45 UTC from IEEE Xplore. Restrictions apply.

as an obstacle detector. It has a wavelength between 600 and 750 nm. To find the direction and distance of objects surrounding the blind, the ESP32 is used. It also operates additional features that notify the user about the type, location, and shape of impediments.

F. Mobile App UI

We have created a web application to provide information about the location and condition of blind individuals. The app can monitor the person and allow guardians to guide them as well. For the front end, we used HTML, CSS, and JavaScript, and Node.js for the backend. Java was utilized to run the app on mobile devices. A server database is capable of holding all the information. In Fig. 8 demonstrates the mobile app UI, which consists of the visual representation.



Fig. 8: Mobile App UI (mobile user interface)

IV. PROGRAMMING AND IMPLEMENTATION

The project flowchart and full system implementation will all be addressed in this section.

A. Flowchart

In Fig. 9 illustrates a flowchart that describes the operation of the Smart Blind Stick. By comparing sensed values to predefined thresholds, this system uses infrared (IR) and ultrasonic sensors to identify obstacles. Once an obstruction is detected, the system classifies it as either water or a physical object. When an obstacle is classified, the device alerts the user in-person by playing a voice alarm that has been pre-recorded. It also provides the blind person's guardian with an SMS that includes the GPS location. Using the MQTT communication protocol, all sensor data are continually transferred to a cloud server, ensuring data accessibility and real-time monitoring. The Smart Blind Stick is a dependable tool for the blind due to its smooth integration of forefront sensors, communication technology, and safety features.

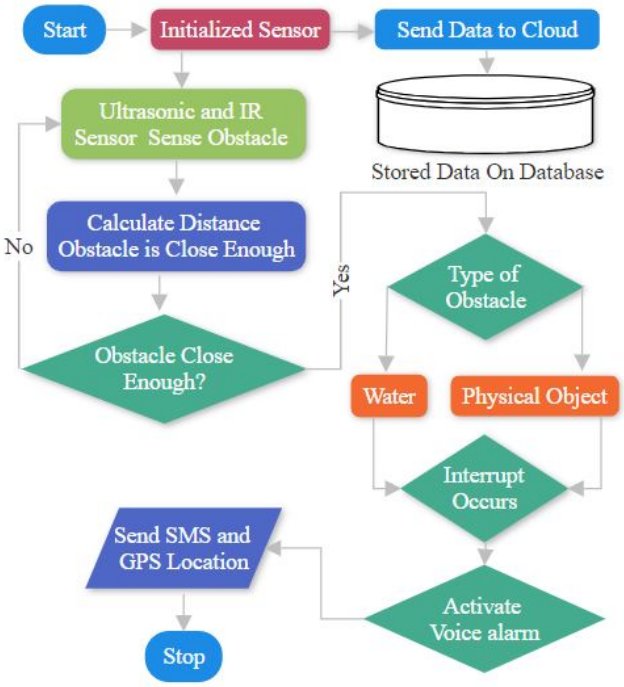


Fig. 9: Flowchart of the Project.

B. Result

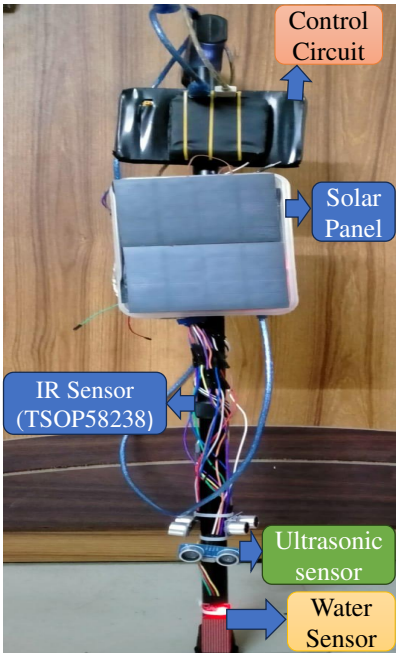


Fig. 10: Overview of the Project.

The prototype implementation of a complete assistance system for visually impaired people is demonstrated in Fig. 10. An ESP32 microcontroller is integrated in this creative

Authorized licensed use limited to: VIT University. Downloaded on October 29,2026 at 03:22:45 UTC from IEEE Xplore. Restrictions apply.

system along with a number of sensors and modules, such as an IR and ultrasonic sensor for obstacle detection, a water level sensor to detect water on the road, and a mini solar panel to charge a Li-ion battery for continuous operation. Guardians receive SMS notifications and real-time location tracking, according to the SIM800L GSM module and NEO-6M GPS. An SD Card Reader also provides voice messages that were previously recorded for instant alerts. The MQTT communication protocol allows data transfer to a cloud server, enabling guardians to keep updated on the person's status via a mobile application or website. With just one click, this platform's simple interface can pinpoint a person's exact location using a marker-based display, guaranteeing immediate assistance and enhanced security.

V. Cost Analysis

Within this portion, we deconstruct the financial assessment for our study project. Based on estimations, the current cost of our prototype is approximately Tk 5647. Considerable cost savings are conceivable if we move to commercial production. By incorporating the components onto a PCB board, the anticipated price would probably drop to about Tk. 4800, ensuring affordability.

TABLE III: EVALUTION OF THE RECOMMENED SYSTEM COST

SL No	Component Name	Quantity	Price (BDT)
1	ESP32	1	380tk
2	HC-SR04 Ultrasonic Sensor	3	279tk
3	GPS module	1	580tk
4	GSM module	1	800tk
5	Water Level Sensor	1	49tk
6	IR Sensor TSOP38238	1	150tk
7	SD Card Reader	1	499tk
8	Mini Solar Panel	1	580tk
9	TP4056 Battery Charging Module	1	65tk
10	Li-ion Battery	2	1665tk
11	Amplifier Module	1	250tk
12	MCP1700 Current Regulator	1	50tk
Total Installation Cost			5347 tk
Miscellaneous Cost			300tk

Nevertheless, a comprehensive study necessitates a full feasibility assessment. "Table II" illustrates the overall hardware costs associated with this study.

Total expense = Estimated Installation Expense + Extra Expense = (5347+300) = 5647 Taka
Which is equivalent to 47.85$ USD.

Conclusion

Our study aims to improve the Smart Blind Stick's performance and intelligence. This System provides visually challenged people real time tracking information regarding their movements. Although all of the information is stored on a cloud server, guardians can access a website or mobile app to keep track on the location. Gloomy weather can impact the effectiveness of long-distance walking. Algorithms for data analysis will be implemented in future updates. All things considered, this method assures an innovative, secure, and effective way to assisting people who are blind.

References

[1] M. M. Rahman, "Blind and visually impaired students in higher education of bangladesh: Barriers and facilitators," Ph.D. dissertation, University of Illinois at Chicago, 2022.

[2] H. Q. Nguyen, A. H. L. Duong, M. D. Vu, T. Q. Dinh, and H. T. Ngo, "Smart blind stick for visually impaired people," in *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries*. Springer, 2022, pp. 145–165.

[3] M. Arunkumar and E. Lokesh, "Iot-based navigation assistance for visually impaired people," in *2023 Eighth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)*. IEEE, 2023, pp. 1–4.

[4] S. Prakash Vats, B. Saha, S. Biswal, R. Kumar, and A. Raj, "Smart stick for visually impaired using iot," *Available at SSRN 4482655*, 2023.

[5] S. Jadon, S. Taluri, S. Birthi, S. Mahesh, S. Kumar, S. S. Shashidhar, and P. B. Honnavalli, "An assistive model for the visually impaired integrating the domains of iot, blockchain and deep learning," *Symmetry*, vol. 15, no. 9, p. 1627, 2023.

[6] M. Javed, Z. Rahman, K. Saad, M. Ashrafi, S. Akter, and A. Rashid, "Design and development of smart blind stick for visually impaired people," in *IOP Conference Series: Materials Science and Engineering*, vol. 1305, no. 1. IOP Publishing, 2024, p. 012032.

[7] Y. Hashim and A. G. Abdulbaqi, "Smart stick for blind people with wireless emergency notification," *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 22, no. 1, pp. 175–181, 2024.

[8] M. A. Khan, M. A. Qureshi, H. Zahid, S. Agha, T. Mushtaq, and S. Nasim, "Innovative blind stick for the visually impaired," *The Asian Bulletin of Big Data Management*, vol. 4, no. 1, 2024.

[9] A. Selvi, V. Jaswant, N. Arunass, R. Arunkumar, and M. K. Prasath, "Design of blind assistive system using internet of things-a survey," in *2024 5th International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*. IEEE, 2024, pp. 680–684.

[10] R. K. Kaushal, T. P. Kumar, N. Sharath, S. Parikh, L. Natrayan, and H. Patil, "Navigating independence: The smart walking stick for the visually impaired," in *2024 5th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*. IEEE, 2024, pp. 103–108.

[11] H. S. Abuelmakarem, A. Abuelhaag, M. Raafat, and S. Ayman, "An integrated iot smart cane for the blind and visually impaired individuals," *SVU-International Journal of Engineering Sciences and Applications*, vol. 5, no. 1, pp. 71–78, 2024.

[12] Z. Khalib, N. Ghazali, M. Elshaikh, and H. Salimin, "Design of smart assist device for the visually impaired," in *AIP Conference Proceedings*, vol. 2898, no. 1. AIP Publishing, 2024.

[13] S. Mohite, M. Patel, A. Patel, and V. Gaikwad, "Smart walking stick for visually impaired," *Computer Science*, vol. 3, pp. 4057–4062, 2024.

[14] K. T. Kahar, K. E. Shingade, S. V. Shah, and T. S. Kale, "Assistive navigational stick," *SSGM Journal of Science and Engineering*, vol. 2, no. 1, pp. 68–73, 2024.

[15] V. R. Pati, P. Kotha, C. L. Kumar M *et al.*, "Navigating the unseen: Ultrasonic technology for blind navigation," 2024.

[16] S. Dhanalakshmi and M. Poongothai, "Embedded assistive cane for visually impaired persons." *Grenze International Journal of Engineering & Technology (GIJET)*, vol. 10, no. 1, 2024.

[17] M. E. Alam, M. A. Kader, U. Hany, R. Arjuman, A. Siddika, and M. Z. Islam, "A multi-controlled semi-autonomous wheelchair for old and physically challenged people," in *2019 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT)*. IEEE, 2019, pp. 1–6.

[18] M. M. Hassain, M. F. U. Mazumder, M. R. Arefin, and M. A. Kader, "Design and implementation of smart head-motion controlled wheelchair," in *2023 IEEE Engineering Informatics*. IEEE, 2023, pp. 1–8.

Authorized licensed use limited to: VIT University. Downloaded on October 29,2026 at 03:22:45 UTC from IEEE Xplore. Restrictions apply.